

haproxy配置peers + stick-table实现多活LB会话持久化

原理概括：

- haproxy将lb当中的amphora加入到同一个peers集群，启用会话持久化的pool会配置一个绑定peers集群的stick-table，当集群中有一个amphora的stick-table更新时，该amphora会将变更同步给该集群中的其他amphora，amphora在收到同步包之后会增量更新自己的stick-table，会话持久化开启时，流量按照stick-table条目中记录的key及server_key负载到对应的member上。

配置示例：

```
1 peers my_peers
2     peer amphora1 192.168.0.101:1025
3     peer amphora2 192.168.0.102:1025
4     peer amphora3 192.168.0.103:1025
5     peer amphora4 192.168.0.104:1025
6
7 backend pool_UUID:listener_UUID
8     stick-table type ipv6 size 10k peers my_peers
```

- 多活模式下要求一个peer集群中所有的amphora必须定义相同的peers列表，名称和顺序也要求必须一致

stick-table配置参数及说明：

配置项	说明	触发更新
type	定义存储key的类型：ip ipv6（Source IP） integer（Session ID） string（COOKIE） binary	无关
size	表大小（entry数量上限），超过会触发LRU淘汰	无关
key	表索引，如IP、COOKIE，只有在key不同时才会有新的记录	新key
use	引用计数，ACL匹配到key并使用时会+1，正常情况下会快速回到0	变化时
exp	过期时间，0表示当前不会过期	新请求重置该字段时
server_id	绑定的后端ID，2表示有两个后端	变化时
server_key	绑定的后端标识，一般为member UUID	变化时
store（会话保持仅使用了server_id和server_key）	定义要在 entry 里存哪些计数器或标记： • gpc0, gpc1 → 通用计数器 • conn_cnt, sess_cnt → 当前连接/会话数 • conn_rate(10s), sess_rate(30s) → 单位时间内的速率 • bytes_in_rate, bytes_out_rate → 吞吐 • http_req_cnt, http_req_rate, http_err_cnt ...	字段值变化时

- server_id, server_key → 会话保持用

SOURCE IP配置示例：

```
1 # table: f479fe20-d255-4966-9d33-885a43c1c5a1:2556f77f-eb37-4487-a51e-28
2 0x56255b9cd6d0: key=:ffff:192.168.100.99 use=0 exp=0 server_id=2 server
```

- 采用ipv6作为key, ipv4客户端地址会被映射成ipv6格式存储

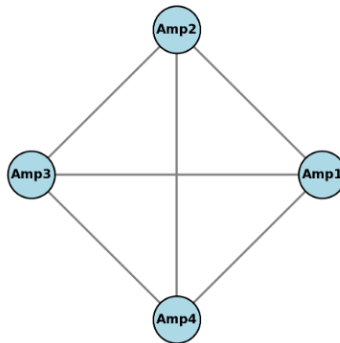
APP COOKIE配置示例：

```
1 # table: f810c239-85ff-40b5-b896-a99166587423:eccfcb30-f725-4072-8bbb-05
2 0x55ea25b031d0: key=member1 use=0 exp=0 server_id=1 server_key=7b0c12ec-
3 0x55ea25becd50: key=member2 use=0 exp=0 server_id=2 server_key=0a9dde4a-
```

- 采用后端定义的APP COOKIE作为key
- 另外：stick-table的更新依赖于 **请求-响应的完整生命周期**。如果一次连接完成了请求（member收到client的包），却没有响应（client没有收到回包），stick-table将不会更新，table当中的used会+1，以显示有请求进入。这种情况在member使用nc监听，且不设置返回值时可以观察到。

stick-table具体更新流程：

- 当peers集群中一个amphora的stick-table更新时，该amphora会通过**单点多播**的方式将更新包发送给peers集群中其他所有的amphora，在其他amphora收到更新包并更新stick-table之后，会返回应答包。
- peers集群中的amphora以**全互联**的方式同步stick-table，在没有stick-table的更新时，会不断的发送心跳包以确保连接
- 更新包内容：不是一整个stick-table，而是stick-table更新的条目，在接收到更新包之后，amphora通过条目中的key对stick-table当中对应的条目进行**覆盖式更新**，stick-table没有这个key，就会新增该条目。



stick-table同步时长：

- 在测试4活lb的peers集群同步stick-table时，有以下抓包结果供参考

72.12.40462	102.100.66.21	102.100.66.100	TCP	234.3825 = 20023 [PWR, ACK] Seq=53 Ack=207 Win=100 Len=148 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第一条数据
73.12.40468	102.100.66.21	102.100.66.100	TCP	234.3825 = 20023 [PWR, ACK] Seq=53 Ack=207 Win=100 Len=148 Tls=1:287000026 Tls=1:287000026	
74.12.40469	102.100.66.21	102.100.66.100	TCP	234.3825 = 20023 [PWR, ACK] Seq=53 Ack=207 Win=100 Len=148 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第二条数据
75.12.40470	102.100.66.21	102.100.66.100	TCP	74.38019 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
80.12.40480	102.100.66.210	102.100.66.41	TCP	74.38019 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第三条数据
82.12.40470	102.100.66.210	102.100.66.41	TCP	74.38019 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
83.12.40487	102.100.66.136	102.100.66.100	HTTP	112.477073.3.100.0	无返回包到amphora的第四条数据
85.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
86.12.47704	102.100.66.100	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第五条数据
87.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
92.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第六条数据
94.12.47704	102.100.66.210	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
95.12.47704	102.100.66.210	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第七条数据
100.12.40480	102.100.66.100	102.100.66.100	HTTP	212.107.1.477073.1	
100.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第八条数据
100.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
107.12.47704	102.100.66.21	102.100.66.100	TCP	82.38021 = 20023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第九条数据
108.12.47704	102.100.66.100	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	
109.12.47704	102.100.66.210	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	无返回包到amphora的第十条数据
112.12.47704	102.100.66.210	102.100.66.41	TCP	74.38021 = 30023 [PWR, ACK] Seq=103 Ack=173 Win=100 Len=8 Tls=1:287000026 Tls=1:287000026	

- 可以看到一次stick-table的更新完成需要2ms左右的时间

stick-table竞态问题：

- 产生条件：负载方式为**轮询的多活lb**开启**Source IP持久化**后，来自于**同一个IP**并发的多个请求，如果stick-table上**不存在以该IP作为key的条目**，那么就会有：本次并发的请求会被**多个amphora负载到pool中所有健康member上**（轮询），导致此次并发的请求在**不同的member上摆动**，stick-table就会**反复更新**条目中的server_id，并向peers集群中的其他amphora发送同步包，直至并发结束
- 结果：根据最后一个请求转发的member更新stick-table，并向其他amphora发送同步包，所有amphora会指向一个member，后续请求（不论是否并发）均会根据stick-table保持持久化
- 影响：业务冲突，导致部分业务丢失或者error，后续业务持久化失败，无法立刻恢复
- 解决方案：划定多活lb的持久化界限：多活情况下只允许配置Cookie类持久化，不允许用户使用Source IP持久化
- 测试如下：

- 服务端：轮询双活lb，配置HTTP监听器和资源池，资源池开启Source IP持久化，有两个健康后端
- 客户端：使用seq 1 10 | xargs -I{} -P10 curl <http://<lb-vip>:<listener-port>>并发10个请求
- 数据获取：amphora用脚本循环追加stick-table到文件中，并在amphora上进行抓包（放在最后）
- 实际数据及现象如下：

- amphora1 (192.168.178.201) 、 amphora2 (192.168.178.50)
- member1/server_id=1 (192.168.178.131) 、 member2/server_id=2 (192.168.178.177)
- amphora1 (192.168.178.201)
- 分析：stick-table第一次变化时used=5，说明有5条请求并发进入，本次记录的是最近一次amphora到member请求的server_id，将其写入stick-table的数据，这里server_id=2；stick-table第二次变化时used=2，但与上次变化相差1s，期间应该stick-table应该有多次更新，第一次并发的请求也早已完成，已经确定了的后端的server_id=1，记录时间差应该是脚本导致的，无法精准记录每次stick-table的变化

```

1 [2025-12-17 16:34:47.195] lb-8d622f7e-3ec7-46a4-9f7f-bcc382700690.novalocal
2 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:0
3
4
5 [2025-12-17 16:34:47.602] lb-8d622f7e-3ec7-46a4-9f7f-bcc382700690.novalocal
6 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:0
7
8
9 [2025-12-17 16:34:48.004] lb-8d622f7e-3ec7-46a4-9f7f-bcc382700690.novalocal
10 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:5
11 0x55c255bf4d80: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=2 server_key=8eabfa6a-5d32-4c84-8c5d-984d592f45c6
12
13
14 [2025-12-17 16:34:49.171] lb-8d622f7e-3ec7-46a4-9f7f-bcc382700690.novalocal
15 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:2
16 0x55c255bf4d80: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=1 server_key=6330a0b9-63ce-4851-a793-a91e8a128a3b
17
18
19 [2025-12-17 16:34:49.761] lb-8d622f7e-3ec7-46a4-9f7f-bcc382700690.novalocal
20 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:2
21 0x55c255bf4d80: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=1 server_key=6330a0b9-63ce-4851-a793-a91e8a128a3b

```

- amphora2 (192.168.178.50)
- 分析：stick-table还未更新时已经显示used=3，说明10次并发，amphora2分到了3个，amphora1分到了7个（可以看抓包结果验证），第一次更新server_id=1，这是该amphora的自发更新，说明最近一次请求应该是到了member1上；第二次更新server_id=2，本次更新时used=4，说明来到了第二次并发，根据member2上的访问记录来看

```

[root@server ~]# python3 simple_server.py
Server started on port 80
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -

```

基本确定这里的server_id=2是被另一个amphora同步过来的，并且根据member2上的结果可以看到，第2次并发的多个请求均有amphora2打到了member2上，最后还是发生了第三次更新，即server_id=1，那此次更新也是被amphora1同步过来的，这点可以由member1上的访问记录验证

```

[root@ulq-um2 ~]# python3 simple_server.py
Server started on port 80
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.50 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -
192.168.178.201 - "GET / HTTP/1.1" 200 -

```

amphora2第1次并发被分配了3个请求，1个到了member1上，2个到了member2上，请求完成要比amphora1快得多，所以stick-table会被amphora1发来的同步包覆盖掉，amphora1第1次并发被分配了7个请求，3个落在member1，4个落在member2，amphora1的请求完成普遍比amphora2慢，所以amphora1的stick-table变化会更慢一步同步到amphora2上，这也就导致了，第二次并发时，amphora2已经根据第二次更新的stick-table向member2发送了5个请求，最终还是被第一次并发结束的amphora1发送的同步结果纠正到了member1上，之后两台amphora的请求只打member1，竞态结束

```
1 [2025-12-17 16:34:47.644] lb-d7da4b40-1390-4922-8665-5f5e780739b8.novalocal
2 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:0
3
4
5 [2025-12-17 16:34:47.815] lb-d7da4b40-1390-4922-8665-5f5e780739b8.novalocal
6 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:3
7
8
9 [2025-12-17 16:34:48.270] lb-d7da4b40-1390-4922-8665-5f5e780739b8.novalocal
10 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:1
11 0x55ad71ade930: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=1 server_key=6330a0b9-63ce-4851-a793-a91e8a128a3b
12
13
14 [2025-12-17 16:34:48.438] lb-d7da4b40-1390-4922-8665-5f5e780739b8.novalocal
15 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:4
16 0x55ad71ade930: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=2 server_key=8eabfa6a-5d32-4c84-8c5d-984d592f45c6
17
18
19 [2025-12-17 16:34:48.843] lb-d7da4b40-1390-4922-8665-5f5e780739b8.novalocal
20 # table: 05e52a40-bb1c-441c-8e2b-d53e91b89e4b:c21dc5f4-a9a3-48d0-970e-987cca74c452, type: ipv6, size:10240, used:1
21 0x55ad71ade930: key::ffff:172.16.10.3 use=0 exp=0 shard=0 server_id=1 server_key=6330a0b9-63ce-4851-a793-a91e8a128a3b
```

性能影响：

- peers集群需要可靠低延迟同步，集群中amphora数量增加所产生的性能影响（待测试）
- 目前多活lb支持最多8台amphora，stick-table peers会话持久化的同步量边界（需进一步调研）
- 同步量较高时对主业务流量的影响（待测试）

参考连接：

peers配置文档：[HAProxy version 3.4-dev1 - Configuration Manual](#)

stick-table配置文档：[Stick tables](#)

amphora1的抓包结果：

```
1 [root@lb-8d622f7e-3ec7-46a4-9ff7f-bcc382700690 ~]# tcpdump -i eth1 -vvv -nne
2 dropped privs to tcpdump
3 tcpdump: listening on eth1, link-type EN10MB (Ethernet), capture size 262144 bytes
4 16:34:44.231152 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 64081, offset 0, flags [
5     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0x8736 (correct), seq 771545555:771545557, ack 3316317732, win 1805,
6     length 2
7 16:34:44.231709 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21276, offset 0, flags [
8     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0xe673 (incorrect -> 0x7f2b), seq 1, ack 2, win 1820, options [nop,nop
9 16:34:44.464967 fa:16:3e:59:62:85 > ff:ff:ff:ff:ff:ff, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-ha
10 78.121, length 28
11 16:34:44.465162 fa:16:3e:59:62:85 > ff:ff:ff:ff:ff:ff, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-ha
12 78.121, length 28
13 16:34:45.177356 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 21277, offset 0, flags [
14     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0xe675 (incorrect -> 0x7b6c), seq 1:3, ack 2, win 1820, options [nop,
15 16:34:45.183560 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64082, offset 0, flags [
16     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0x77d2 (correct), seq 2, ack 3, win 1805, options [nop,nop,TS val 2025
17 16:34:47.231873 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 64083, offset 0, flags [
18     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0x6fc0 (correct), seq 2:4, ack 3, win 1805, options [nop,nop,TS val 2
19 16:34:47.232135 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21278, offset 0, flags [
20     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0xe673 (incorrect -> 0x67b6), seq 3, ack 4, win 1820, options [nop,nop
21 16:34:48.135706 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 7045, offset 0, flags [D
22     172.16.10.3.42680 > 192.168.178.40.80: Flags [S], cksum 0xaaaf (correct), seq 992147678, win 29200, options [mss 1460,sackOK,TS val
23 16:34:48.140436 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
24     192.168.178.40.80 > 172.16.10.3.42680: Flags [S.], cksum 0x2913 (incorrect -> 0x4390), seq 299127546, ack 992147679, win 27800, opti
25 4,nop,wscale 4], length 0
26 16:34:48.141187 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 47732, offset 0, flags [
27     172.16.10.3.42682 > 192.168.178.40.80: Flags [S], cksum 0xba32 (correct), seq 2992468590, win 29200, options [mss 1460,sackOK,TS val
28 16:34:48.141391 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
29     192.168.178.40.80 > 172.16.10.3.42682: Flags [S.], cksum 0x2913 (incorrect -> 0x43d5), seq 3523649972, ack 2992468591, win 27800, op
30 124,nop,wscale 4], length 0
31 16:34:48.141610 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 5708, offset 0, flags [D
32     172.16.10.3.42690 > 192.168.178.40.80: Flags [S], cksum 0x8ab1 (correct), seq 2521546233, win 29200, options [mss 1460,sackOK,TS val
33 16:34:48.141769 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
34     192.168.178.40.80 > 172.16.10.3.42690: Flags [S.], cksum 0x2913 (incorrect -> 0xb068), seq 1565358680, ack 2521546234, win 27800, op
35 124,nop,wscale 4], length 0
36 16:34:48.141826 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 20973, offset 0, flags [
37     172.16.10.3.42700 > 192.168.178.40.80: Flags [S], cksum 0x9db9 (correct), seq 3342367225, win 29200, options [mss 1460,sackOK,TS val
38 16:34:48.141952 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
39     192.168.178.40.80 > 172.16.10.3.42700: Flags [S.], cksum 0x2913 (incorrect -> 0x5604), seq 3666242187, ack 3342367226, win 27800, op
40 125,nop,wscale 4], length 0
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41 16:34:48.141996 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 49060, offset 0, flags [
42 172.16.10.3.42704 > 192.168.178.40.80: Flags [S], cksum 0x8b25 (correct), seq 3620633586, win 29200, options [mss 1460,sackOK,TS val
43 16:34:48.142106 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
44 192.168.178.40.80 > 172.16.10.3.42704: Flags [S.], cksum 0x2913 (incorrect -> 0xbf3c), seq 3964394745, ack 3620633587, win 27800, op
45 126,nop,wscale 4], length 0
46 16:34:48.142141 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 31658, offset 0, flags [
47 172.16.10.3.42718 > 192.168.178.40.80: Flags [S], cksum 0x77cb (correct), seq 428428674, win 29200, options [mss 1460,sackOK,TS val
48 16:34:48.142250 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
49 192.168.178.40.80 > 172.16.10.3.42718: Flags [S.], cksum 0x2913 (incorrect -> 0x4c21), seq 1744158993, ack 428428675, win 27800, opt
50 28,nop,wscale 4], length 0
51 16:34:48.142290 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 5697, offset 0, flags [D
52 172.16.10.3.42724 > 192.168.178.40.80: Flags [S], cksum 0xa3be (correct), seq 479862391, win 29200, options [mss 1460,sackOK,TS val
53 16:34:48.142428 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
54 192.168.178.40.80 > 172.16.10.3.42724: Flags [S.], cksum 0x2913 (incorrect -> 0x832e), seq 2293339451, ack 479862392, win 27800, opt
55 29,nop,wscale 4], length 0
56 16:34:48.142809 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 7047, offset 0, flags [
57 172.16.10.3.42680 > 192.168.178.40.80: Flags [P.], cksum 0xa5ee (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
58 GET / HTTP/1.1
59 Host: 172.16.10.21
60 User-Agent: curl/7.61.1
61 Accept: /*/*
62
63 16:34:48.148273 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 7046, offset 0, flags [D
64 172.16.10.3.42680 > 192.168.178.40.80: Flags [.], cksum 0xde76 (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730131 e
65 16:34:48.148589 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 47733, offset 0, flags [
66 172.16.10.3.42682 > 192.168.178.40.80: Flags [.], cksum 0xdbeb (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730131 e
67 16:34:48.148848 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 47734, offset 0, flags
68 172.16.10.3.42682 > 192.168.178.40.80: Flags [P.], cksum 0xa632 (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
69 GET / HTTP/1.1
70 Host: 172.16.10.21
71 User-Agent: curl/7.61.1
72 Accept: /*/*
73
74 16:34:48.148897 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5709, offset 0, flags [D
75 172.16.10.3.42690 > 192.168.178.40.80: Flags [.], cksum 0x4b4e (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730132 e
76 16:34:48.148997 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 5710, offset 0, flags [
77 172.16.10.3.42690 > 192.168.178.40.80: Flags [P.], cksum 0x12c6 (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
78 GET / HTTP/1.1
79 Host: 172.16.10.21
80 User-Agent: curl/7.61.1
81 Accept: /*/*
82
83 16:34:48.149030 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 20975, offset 0, flags
84 172.16.10.3.42700 > 192.168.178.40.80: Flags [P.], cksum 0xb862 (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
85 GET / HTTP/1.1
86 Host: 172.16.10.21
87 User-Agent: curl/7.61.1
88 Accept: /*/*
89
90 16:34:48.149119 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 20974, offset 0, flags [
91 172.16.10.3.42700 > 192.168.178.40.80: Flags [.], cksum 0xf0ea (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730132 e
92 16:34:48.149145 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 49061, offset 0, flags [
93 172.16.10.3.42704 > 192.168.178.40.80: Flags [.], cksum 0x5a24 (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730132 e
94 16:34:48.149241 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 49062, offset 0, flags
95 172.16.10.3.42704 > 192.168.178.40.80: Flags [P.], cksum 0x219c (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
96 GET / HTTP/1.1
97 Host: 172.16.10.21
98 User-Agent: curl/7.61.1
99 Accept: /*/*
100
101 16:34:48.149271 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 31659, offset 0, flags [
102 172.16.10.3.42718 > 192.168.178.40.80: Flags [.], cksum 0xe70a (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730132 e
103 16:34:48.149377 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 31660, offset 0, flags
104 172.16.10.3.42718 > 192.168.178.40.80: Flags [P.], cksum 0xae82 (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
105 GET / HTTP/1.1
106 Host: 172.16.10.21
107 User-Agent: curl/7.61.1
108 Accept: /*/*
109
110 16:34:48.149403 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5698, offset 0, flags [D
111 172.16.10.3.42724 > 192.168.178.40.80: Flags [.], cksum 0x1e19 (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730132 e
112 16:34:48.149577 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 5699, offset 0, flags [
113 172.16.10.3.42724 > 192.168.178.40.80: Flags [P.], cksum 0xe58f (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
114 GET / HTTP/1.1
115 Host: 172.16.10.21
116 User-Agent: curl/7.61.1
117 Accept: /*/*
118
119 16:34:48.189647 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 53274, offset 0, flags [
120 192.168.178.40.80 > 172.16.10.3.42680: Flags [.], cksum 0x290b (incorrect -> 0xd769), seq 1, ack 77, win 1738, options [nop,nop,TS v
121 16:34:48.189947 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 17500, offset 0, flags [
122 192.168.178.40.80 > 172.16.10.3.42718: Flags [.], cksum 0x290b (incorrect -> 0xdffe), seq 1, ack 77, win 1738, options [nop,nop,TS v
123 16:34:48.190001 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31484, offset 0, flags [
```



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124 192.168.178.40.80 > 172.16.10.3.42704: Flags [.] , cksum 0x290b (incorrect -> 0x5318), seq 1, ack 77, win 1738, options [nop,nop,TS v
125 16:34:48.190089 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63756, offset 0, flags [
126 192.168.178.40.80 > 172.16.10.3.42700: Flags [.] , cksum 0x290b (incorrect -> 0xe9de), seq 1, ack 77, win 1738, options [nop,nop,TS v
127 16:34:48.190131 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37026, offset 0, flags [
128 192.168.178.40.80 > 172.16.10.3.42690: Flags [.] , cksum 0x290b (incorrect -> 0x4442), seq 1, ack 77, win 1738, options [nop,nop,TS v
129 16:34:48.190197 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 56389, offset 0, flags [
130 192.168.178.40.80 > 172.16.10.3.42682: Flags [.] , cksum 0x290b (incorrect -> 0xd7ad), seq 1, ack 77, win 1738, options [nop,nop,TS v
131 16:34:48.191527 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31894, offset 0, flags [
132 192.168.178.40.80 > 172.16.10.3.42724: Flags [.] , cksum 0x290b (incorrect -> 0x170b), seq 1, ack 77, win 1738, options [nop,nop,TS v
133 16:34:48.284824 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 28779, offset 0, flags [
134 192.168.178.201.4735 > 192.168.178.177.80: Flags [S], cksum 0xe6fa (incorrect -> 0x4b52), seq 3027999506, win 28040, options [mss 14
135 gth 0
136 16:34:48.286727 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
137 192.168.178.177.80 > 192.168.178.201.4735: Flags [S.], cksum 0xb6e8 (correct), seq 2082056536, ack 3027999507, win 27800, options [m
138 scale 4], length 0
139 16:34:48.287887 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 28780, offset 0, flags [
140 192.168.178.201.4735 > 192.168.178.177.80: Flags [.] , cksum 0xe6f2 (incorrect -> 0x4b33), seq 1, ack 1, win 1753, options [nop,nop,T
141 16:34:48.304866 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 7246, offset 0, flags [D
142 192.168.178.201.58439 > 192.168.178.131.80: Flags [S], cksum 0xe6cc (incorrect -> 0x9431), seq 857827356, win 28040, options [mss 14
143 gth 0
144 16:34:48.308055 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
145 192.168.178.131.80 > 192.168.178.201.58439: Flags [S.], cksum 0xf629 (correct), seq 431291831, ack 857827357, win 27800, options [ms
146 scale 4], length 0
147 16:34:48.308335 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 7247, offset 0, flags [D
148 192.168.178.201.58439 > 192.168.178.131.80: Flags [.] , cksum 0xe6c4 (incorrect -> 0x8a75), seq 1, ack 1, win 1753, options [nop,nop,T
149 16:34:48.315270 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 14941, offset 0, flags [
150 192.168.178.201.4747 > 192.168.178.177.80: Flags [S], cksum 0xe6fa (incorrect -> 0xef6b), seq 2449414474, win 28040, options [mss 14
151 gth 0
152 16:34:48.318128 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
153 192.168.178.177.80 > 192.168.178.201.4747: Flags [S.], cksum 0xe6fa (incorrect -> 0x9ab2), seq 665959920, ack 2449414475, win 27800,
154 16868,nop,wscale 4], length 0
155 16:34:48.318873 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 14942, offset 0, flags [
156 192.168.178.201.4747 > 192.168.178.177.80: Flags [.] , cksum 0xe6f2 (incorrect -> 0x2efd), seq 1, ack 1, win 1753, options [nop,nop,T
157 16:34:48.323200 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 2015, offset 0, flags [D
158 192.168.178.201.58447 > 192.168.178.131.80: Flags [S], cksum 0xe6cc (incorrect -> 0xa880), seq 2869746631, win 28040, options [mss 1
159 ngth 0
160 16:34:48.324571 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
161 192.168.178.131.80 > 192.168.178.201.58447: Flags [S.], cksum 0x8fd8 (correct), seq 1061704371, ack 2869746632, win 27800, options [
162 ,wscale 4], length 0
163 16:34:48.324837 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 2016, offset 0, flags [D
164 192.168.178.201.58447 > 192.168.178.131.80: Flags [.] , cksum 0xe6c4 (incorrect -> 0x2425), seq 1, ack 1, win 1753, options [nop,nop,
165 16:34:48.362520 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 64084, offset 0, flags [
166 192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xfbf3 (correct), seq 4:30, ack 3, win 1805, options [nop,nop,TS val
167 16:34:48.362934 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21279, offset 0, flags [
168 192.168.178.201.49233 > 192.168.178.50.1025: Flags [.] , cksum 0xe673 (incorrect -> 0x5ec6), seq 3, ack 30, win 1820, options [nop,no
169 16:34:48.375036 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 28781, offset 0, flags
170 192.168.178.201.4735 > 192.168.178.177.80: Flags [P.], cksum 0xe75c (incorrect -> 0x5602), seq 1:107, ack 1, win 1753, options [nop,
171 TP, length: 106
172 GET / HTTP/1.1
173 host: 172.16.10.21
174 user-agent: curl/7.61.1
175 accept: /*
176 x-forwarded-for: 172.16.10.3
177
178 16:34:48.375834 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 51144, offset 0, flags [
179 192.168.178.177.80 > 192.168.178.201.4735: Flags [.] , cksum 0xe6f2 (incorrect -> 0x4a28), seq 1, ack 107, win 1738, options [nop,nop
180 16:34:48.383443 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 7248, offset 0, flags [
181 192.168.178.201.58439 > 192.168.178.131.80: Flags [P.], cksum 0xe72e (incorrect -> 0x9550), seq 1:107, ack 1, win 1753, options [nop
182 HTTP, length: 106
183 GET / HTTP/1.1
184 host: 172.16.10.21
185 user-agent: curl/7.61.1
186 accept: /*
187 x-forwarded-for: 172.16.10.3
188
189 16:34:48.385720 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 9, offset 0, flags [DF],
190 192.168.178.131.80 > 192.168.178.201.58439: Flags [.] , cksum 0x8981 (correct), seq 1, ack 107, win 1738, options [nop,nop,TS val 107
191 16:34:48.399555 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 51145, offset 0, flags
192 192.168.178.177.80 > 192.168.178.201.4735: Flags [P.], cksum 0xe767 (incorrect -> 0x1c0c), seq 1:118, ack 107, win 1738, options [no
193 HTTP, length: 117
194 HTTP/1.0 200 OK
195 Server: BaseHTTP/0.6 Python/3.6.8
196 Date: Wed, 17 Dec 2025 08:34:47 GMT
197 Content-Type: text/plain
198
199 16:34:48.399984 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 28782, offset 0, flags [
200 192.168.178.201.4735 > 192.168.178.177.80: Flags [.] , cksum 0xe6f2 (incorrect -> 0x497b), seq 107, ack 118, win 1753, options [nop,n
201 16:34:48.400186 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 51146, offset 0, flags [
202 192.168.178.177.80 > 192.168.178.201.4735: Flags [FP.], cksum 0xe6f7 (incorrect -> 0x9bf5), seq 118:123, ack 107, win 1738, options
203 HTTP
204 16:34:48.401428 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 51147, offset 0, flags [
205 192.168.178.177.80 > 192.168.178.201.4735: Flags [FP.], cksum 0xe6f7 (incorrect -> 0x9bed), seq 118:123, ack 107, win 1738, options
206 HTTP
```

```
207 16:34:48.402040 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 78: (tos 0x0, ttl 64, id 28783, offset 0, flags [
208     192.168.178.201.4735 > 192.168.178.177.80: Flags [.), cksum 0xe6fe (incorrect -> 0xc77d), seq 107, ack 124, win 1753, options [nop,n
209     18:124], length 0
210 16:34:48.407774 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 14943, offset 0, flags
211     192.168.178.201.4747 > 192.168.178.177.80: Flags [P.], cksum 0xe75c (incorrect -> 0x39ca), seq 1:107, ack 1, win 1753, options [nop,
212     TP, length: 106
213         GET / HTTP/1.1
214         host: 172.16.10.21
215         user-agent: curl/7.61.1
216         accept: */*
217         x-forwarded-for: 172.16.10.3
218
219 16:34:48.408065 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 2017, offset 0, flags [
220     192.168.178.201.58447 > 192.168.178.131.80: Flags [P.], cksum 0xe72e (incorrect -> 0x2ef8), seq 1:107, ack 1, win 1753, options [nop
221     HTTP, length: 106
222         GET / HTTP/1.1
223         host: 172.16.10.21
224         user-agent: curl/7.61.1
225         accept: */*
226         x-forwarded-for: 172.16.10.3
227
228 16:34:48.408365 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 9677, offset 0, flags [D
229     192.168.178.177.80 > 192.168.178.201.4747: Flags [.), cksum 0xe6f2 (incorrect -> 0x2def), seq 1, ack 107, win 1738, options [nop,nop
230 16:34:48.409357 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 51106, offset 0, flags [
231     192.168.178.131.80 > 192.168.178.201.58447: Flags [.), cksum 0x2322 (correct), seq 1, ack 107, win 1738, options [nop,nop,TS val 107
232 16:34:48.410444 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 10, offset 0, flags [DF
233     192.168.178.131.80 > 192.168.178.201.58439: Flags [P.], cksum 0x5b5c (correct), seq 1:118, ack 107, win 1738, options [nop,nop,TS va
234     th: 117
235         HTTP/1.0 200 OK
236         Server: BaseHTTP/0.6 Python/3.6.8
237         Date: Wed, 17 Dec 2025 08:34:47 GMT
238         Content-Type: text/plain
239
240 16:34:48.410714 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 7249, offset 0, flags [D
241     192.168.178.201.58439 > 192.168.178.131.80: Flags [.), cksum 0xe6c4 (incorrect -> 0x88c8), seq 107, ack 118, win 1753, options [nop,
242 16:34:48.415864 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 4358, offset 0, flags [D
243     192.168.178.201.4759 > 192.168.178.177.80: Flags [S], cksum 0xe6fa (incorrect -> 0x86d4), seq 1953079046, win 28040, options [mss 14
244     gth 0
245 16:34:48.417193 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 15417, offset 0, flags [
246     192.168.178.201.58449 > 192.168.178.131.80: Flags [S], cksum 0xe6cc (incorrect -> 0x585a), seq 3372420503, win 28040, options [mss 1
247     ngth 0
248 16:34:48.417567 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 40487, offset 0, flags [
249     192.168.178.201.4769 > 192.168.178.177.80: Flags [S], cksum 0xe6fa (incorrect -> 0xf4d2), seq 3910057046, win 28040, options [mss 14
250     gth 0
251 16:34:48.418127 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
252     192.168.178.177.80 > 192.168.178.201.4759: Flags [S.], cksum 0xe6fa (incorrect -> 0x4d17), seq 3818776229, ack 1953079047, win 27800
253     616969,nop,wscale 4], length 0
254 16:34:48.418311 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 4359, offset 0, flags [D
255     192.168.178.201.4759 > 192.168.178.177.80: Flags [.), cksum 0xe6f2 (incorrect -> 0xe163), seq 1, ack 1, win 1753, options [nop,nop,T
256 16:34:48.418940 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
257     192.168.178.177.80 > 192.168.178.201.4769: Flags [S.], cksum 0xe6fa (incorrect -> 0x1b23), seq 3360924127, ack 3910057047, win 27800
258     616971,nop,wscale 4], length 0
259 16:34:48.419744 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 40488, offset 0, flags [
260     192.168.178.201.4769 > 192.168.178.177.80: Flags [.), cksum 0xe6f2 (incorrect -> 0xaf6f), seq 1, ack 1, win 1753, options [nop,nop,T
261 16:34:48.419807 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
262     192.168.178.131.80 > 192.168.178.201.58449: Flags [S.], cksum 0x1b92 (correct), seq 1140748734, ack 3372420504, win 27800, options [
263     ,wscale 4], length 0
264 16:34:48.419924 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 15418, offset 0, flags [
265     192.168.178.201.58449 > 192.168.178.131.80: Flags [.), cksum 0xe6c4 (incorrect -> 0xafdd), seq 1, ack 1, win 1753, options [nop,nop,
266 16:34:48.421927 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 9678, offset 0, flags [
267     192.168.178.177.80 > 192.168.178.201.4747: Flags [P.], cksum 0xe767 (incorrect -> 0xfed5), seq 1:118, ack 107, win 1738, options [no
268     HTTP, length: 117
269         HTTP/1.0 200 OK
270         Server: BaseHTTP/0.6 Python/3.6.8
271         Date: Wed, 17 Dec 2025 08:34:48 GMT
272         Content-Type: text/plain
273
274 16:34:48.422376 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 14944, offset 0, flags [
275     192.168.178.201.4747 > 192.168.178.177.80: Flags [.), cksum 0xe6f2 (incorrect -> 0x2d50), seq 107, ack 118, win 1753, options [nop,n
276 16:34:48.422453 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 9679, offset 0, flags [D
277     192.168.178.177.80 > 192.168.178.201.4747: Flags [FP.], cksum 0xe6f7 (incorrect -> 0x7fbe), seq 118:123, ack 107, win 1738, options
278     HTTP
279 16:34:48.432108 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 11, offset 0, flags [DF]
280     192.168.178.131.80 > 192.168.178.201.58439: Flags [P.], cksum 0xdb16 (correct), seq 118:123, ack 107, win 1738, options [nop,nop,TS
281 16:34:48.432262 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 7250, offset 0, flags [D
282     192.168.178.201.58439 > 192.168.178.131.80: Flags [.), cksum 0xe6c4 (incorrect -> 0x8899), seq 107, ack 123, win 1753, options [nop,
283 16:34:48.432361 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 21280, offset 0, flags [
284     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0xe675 (incorrect -> 0x5e73), seq 3:5, ack 30, win 1820, options [nop
285 16:34:48.432812 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 12, offset 0, flags [DF]
286     192.168.178.131.80 > 192.168.178.201.58439: Flags [F.], cksum 0x88bb (correct), seq 123, ack 107, win 1738, options [nop,nop,TS val
287 16:34:48.434001 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64085, offset 0, flags [
288     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.), cksum 0x5e47 (correct), seq 30, ack 5, win 1805, options [nop,nop,TS val 202
289 16:34:48.451713 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 4360, offset 0, flags [
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290 192.168.178.201.4759 > 192.168.178.177.80: Flags [P.], cksum 0xe75c (incorrect -> 0xec67), seq 1:107, ack 1, win 1753, options [nop,
291 TP, length: 106
292 GET / HTTP/1.1
293 host: 172.16.10.21
294 user-agent: curl/7.61.1
295 accept: */*
296 x-forwarded-for: 172.16.10.3
297
298 16:34:48.451952 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 15419, offset 0, flags
299 192.168.178.201.58449 > 192.168.178.131.80: Flags [P.], cksum 0xe72e (incorrect -> 0xbae3), seq 1:107, ack 1, win 1753, options [nop,
300 HTTP, length: 106
301 GET / HTTP/1.1
302 host: 172.16.10.21
303 user-agent: curl/7.61.1
304 accept: */*
305 x-forwarded-for: 172.16.10.3
306
307 16:34:48.452047 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 40489, offset 0, flags
308 192.168.178.201.4769 > 192.168.178.177.80: Flags [P.], cksum 0xe75c (incorrect -> 0xba75), seq 1:107, ack 1, win 1753, options [nop,
309 TP, length: 106
310 GET / HTTP/1.1
311 host: 172.16.10.21
312 user-agent: curl/7.61.1
313 accept: */*
314 x-forwarded-for: 172.16.10.3
315
316 16:34:48.453178 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63193, offset 0, flags [
317 192.168.178.177.80 > 192.168.178.201.4759: Flags [P.], cksum 0xe6f2 (incorrect -> 0xe0c1), seq 1, ack 107, win 1738, options [nop,nop
318 16:34:48.453367 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 6690, offset 0, flags [D
319 192.168.178.177.80 > 192.168.178.201.4769: Flags [P.], cksum 0xe6f2 (incorrect -> 0xaed2), seq 1, ack 107, win 1738, options [nop,nop,TS
320 16:34:48.453776 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 32325, offset 0, flags [
321 192.168.178.131.80 > 192.168.178.201.58449: Flags [P.], cksum 0xaf40 (correct), seq 1, ack 107, win 1738, options [nop,nop,TS val 107
322 16:34:48.456599 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 51107, offset 0, flags
323 192.168.178.131.80 > 192.168.178.201.58447: Flags [P.], cksum 0xf4e6 (correct), seq 1:118, ack 107, win 1738, options [nop,nop,TS va
324 th: 117
325 HTTP/1.0 200 OK
326 Server: BaseHTTP/0.6 Python/3.6.8
327 Date: Wed, 17 Dec 2025 08:34:47 GMT
328 Content-Type: text/plain
329
330 16:34:48.456787 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 2018, offset 0, flags [D
331 192.168.178.201.58447 > 192.168.178.131.80: Flags [P.], cksum 0xe6c4 (incorrect -> 0x223e), seq 107, ack 118, win 1753, options [nop,
332 16:34:48.456987 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 51108, offset 0, flags [
333 192.168.178.131.80 > 192.168.178.201.58447: Flags [FP.], cksum 0x74d1 (correct), seq 118:123, ack 107, win 1738, options [nop,nop,TS
334 16:34:48.459435 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 6691, offset 0, flags [
335 192.168.178.177.80 > 192.168.178.201.4769: Flags [P.], cksum 0xe767 (incorrect -> 0x7fc0), seq 1:118, ack 107, win 1738, options [no
336 HTTP, length: 117
337 HTTP/1.0 200 OK
338 Server: BaseHTTP/0.6 Python/3.6.8
339 Date: Wed, 17 Dec 2025 08:34:48 GMT
340 Content-Type: text/plain
341
342 16:34:48.459716 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 40490, offset 0, flags [
343 192.168.178.201.4769 > 192.168.178.177.80: Flags [P.], cksum 0xe6f2 (incorrect -> 0xae40), seq 107, ack 118, win 1753, options [nop,n
344 16:34:48.459988 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 6692, offset 0, flags [D
345 192.168.178.177.80 > 192.168.178.201.4769: Flags [FP.], cksum 0xe6f7 (incorrect -> 0x00a9), seq 118:123, ack 107, win 1738, options
346 HTTP
347 16:34:48.464567 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 14945, offset 0, flags [
348 192.168.178.201.4747 > 192.168.178.177.80: Flags [P.], cksum 0xe6f2 (incorrect -> 0x2d1f), seq 107, ack 124, win 1753, options [nop,n
349 16:34:48.467822 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 63194, offset 0, flags
350 192.168.178.177.80 > 192.168.178.201.4759: Flags [P.], cksum 0xe767 (incorrect -> 0xb1a7), seq 1:118, ack 107, win 1738, options [no
351 HTTP, length: 117
352 HTTP/1.0 200 OK
353 Server: BaseHTTP/0.6 Python/3.6.8
354 Date: Wed, 17 Dec 2025 08:34:48 GMT
355 Content-Type: text/plain
356
357 16:34:48.468267 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 4361, offset 0, flags [D
358 192.168.178.201.4759 > 192.168.178.177.80: Flags [P.], cksum 0xe6f2 (incorrect -> 0xe01f), seq 107, ack 118, win 1753, options [nop,n
359 16:34:48.468416 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 63195, offset 0, flags [
360 192.168.178.177.80 > 192.168.178.201.4759: Flags [FP.], cksum 0xe6f7 (incorrect -> 0x3290), seq 118:123, ack 107, win 1738, options
361 HTTP
362 16:34:48.474690 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 7251, offset 0, flags [D
363 192.168.178.201.58439 > 192.168.178.131.80: Flags [P.], cksum 0xe6c4 (incorrect -> 0x886c), seq 107, ack 124, win 1753, options [nop,
364 16:34:48.481319 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 32326, offset 0, flags
365 192.168.178.131.80 > 192.168.178.201.58449: Flags [P.], cksum 0x8119 (correct), seq 1:118, ack 107, win 1738, options [nop,nop,TS va
366 th: 117
367 HTTP/1.0 200 OK
368 Server: BaseHTTP/0.6 Python/3.6.8
369 Date: Wed, 17 Dec 2025 08:34:47 GMT
370 Content-Type: text/plain
371
372 16:34:48.481664 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 15420, offset 0, flags [
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373 192.168.178.201.58449 > 192.168.178.131.80: Flags [..], cksum 0xe6c4 (incorrect -> 0xae83), seq 107, ack 118, win 1753, options [nop,
374 16:34:48.483982 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 32327, offset 0, flags [
375 192.168.178.131.80 > 192.168.178.201.58449: Flags [FP..], cksum 0x0103 (correct), seq 118:123, ack 107, win 1738, options [nop,nop,TS
376 16:34:48.494812 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 64086, offset 0, flags [
377 192.168.178.50.1025 > 192.168.178.201.49233: Flags [P..], cksum 0xf5ad (correct), seq 30:56, ack 5, win 1805, options [nop,nop,TS val
378 16:34:48.495648 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21281, offset 0, flags [
379 192.168.178.201.49233 > 192.168.178.50.1025: Flags [..], cksum 0xe673 (incorrect -> 0x5da2), seq 5, ack 56, win 1820, options [nop,no
380 16:34:48.497744 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 2019, offset 0, flags [D
381 192.168.178.201.58447 > 192.168.178.131.80: Flags [..], cksum 0xe6c4 (incorrect -> 0x220f), seq 107, ack 124, win 1753, options [nop,
382 16:34:48.501704 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 40491, offset 0, flags [
383 192.168.178.201.4769 > 192.168.178.177.80: Flags [..], cksum 0xe6f2 (incorrect -> 0xae0f), seq 107, ack 124, win 1753, options [nop,n
384 16:34:48.501805 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 21282, offset 0, flags [
385 192.168.178.201.49233 > 192.168.178.50.1025: Flags [P..], cksum 0xe67b (incorrect -> 0x4bd5), seq 5:13, ack 56, win 1820, options [no
386 16:34:48.502880 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64087, offset 0, flags [
387 192.168.178.50.1025 > 192.168.178.201.49233: Flags [..], cksum 0x5d9a (correct), seq 56, ack 13, win 1805, options [nop,nop,TS val 20
388 16:34:48.509646 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 4362, offset 0, flags [D
389 192.168.178.201.4759 > 192.168.178.177.80: Flags [..], cksum 0xe6f2 (incorrect -> 0xdfee), seq 107, ack 124, win 1753, options [nop,n
390 16:34:48.525123 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 15421, offset 0, flags [
391 192.168.178.201.58449 > 192.168.178.131.80: Flags [..], cksum 0xe6c4 (incorrect -> 0xae51), seq 107, ack 124, win 1753, options [nop,
392 16:34:48.534605 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 53275, offset 0, flags
393 192.168.178.40.80 > 172.16.10.3.42680: Flags [P..], cksum 0x2985 (incorrect -> 0xe7f1), seq 1:123, ack 77, win 1738, options [nop,nop
394 , length: 122
395 HTTP/1.0 200 OK
396 server: BaseHTTP/0.6 Python/3.6.8
397 date: Wed, 17 Dec 2025 08:34:47 GMT
398 content-type: text/plain
399
400 vm-2
401 16:34:48.535203 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 7048, offset 0, flags [D
402 172.16.10.3.42680 > 192.168.178.40.80: Flags [..], cksum 0xda9c (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373052
403 16:34:48.544736 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 53276, offset 0, flags [
404 192.168.178.40.80 > 172.16.10.3.42680: Flags [F..], cksum 0x290b (incorrect -> 0xd400), seq 123, ack 77, win 1738, options [nop,nop,T
405 16:34:48.545342 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 7049, offset 0, flags [D
406 172.16.10.3.42680 > 192.168.178.40.80: Flags [F..], cksum 0xda85 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537305
407 16:34:48.545793 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 53277, offset 0, flags [
408 192.168.178.40.80 > 172.16.10.3.42680: Flags [..], cksum 0x290b (incorrect -> 0xd3f4), seq 124, ack 78, win 1738, options [nop,nop,TS
409 16:34:48.622421 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 28784, offset 0, flags [
410 192.168.178.201.4735 > 192.168.178.177.80: Flags [F..], cksum 0xe6f2 (incorrect -> 0x488e), seq 107, ack 124, win 1753, options [nop,
411 16:34:48.623946 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
412 192.168.178.177.80 > 192.168.178.201.4735: Flags [..], cksum 0x47bd (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 49
413 16:34:48.637099 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 56390, offset 0, flags
414 192.168.178.40.80 > 172.16.10.3.42682: Flags [P..], cksum 0x2985 (incorrect -> 0xe8cf), seq 1:123, ack 77, win 1738, options [nop,nop
415 , length: 122
416 HTTP/1.0 200 OK
417 server: BaseHTTP/0.6 Python/3.6.8
418 date: Wed, 17 Dec 2025 08:34:47 GMT
419 content-type: text/plain
420
421 vm-1
422 16:34:48.637755 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 47735, offset 0, flags [
423 172.16.10.3.42682 > 192.168.178.40.80: Flags [..], cksum 0xda15 (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373062
424 16:34:48.640847 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 56391, offset 0, flags [
425 192.168.178.40.80 > 172.16.10.3.42682: Flags [F..], cksum 0x290b (incorrect -> 0xd380), seq 123, ack 77, win 1738, options [nop,nop,T
426 16:34:48.641596 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 47736, offset 0, flags [
427 172.16.10.3.42682 > 192.168.178.40.80: Flags [F..], cksum 0xda0b (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537306
428 16:34:48.641826 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 56392, offset 0, flags [
429 192.168.178.40.80 > 172.16.10.3.42682: Flags [..], cksum 0x290b (incorrect -> 0xd37a), seq 124, ack 78, win 1738, options [nop,nop,TS
430 16:34:48.652311 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 7252, offset 0, flags [D
431 192.168.178.201.58439 > 192.168.178.131.80: Flags [F..], cksum 0xe6c4 (incorrect -> 0x87ba), seq 107, ack 124, win 1753, options [nop
432 16:34:48.653663 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
433 192.168.178.131.80 > 192.168.178.201.58439: Flags [..], cksum 0x86ec (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 1
434 16:34:48.657822 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 37027, offset 0, flags
435 192.168.178.40.80 > 172.16.10.3.42690: Flags [P..], cksum 0x2985 (incorrect -> 0x534f), seq 1:123, ack 77, win 1738, options [nop,nop
436 , length: 122
437 HTTP/1.0 200 OK
438 server: BaseHTTP/0.6 Python/3.6.8
439 date: Wed, 17 Dec 2025 08:34:48 GMT
440 content-type: text/plain
441
442 vm-2
443 16:34:48.658331 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5711, offset 0, flags [D
444 172.16.10.3.42690 > 192.168.178.40.80: Flags [..], cksum 0x4680 (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373064
445 16:34:48.661233 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37028, offset 0, flags [
446 192.168.178.40.80 > 172.16.10.3.42690: Flags [F..], cksum 0x290b (incorrect -> 0x3fec), seq 123, ack 77, win 1738, options [nop,nop,T
447 16:34:48.661787 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5712, offset 0, flags [D
448 172.16.10.3.42690 > 192.168.178.40.80: Flags [F..], cksum 0x4678 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537306
449 16:34:48.662587 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37029, offset 0, flags [
450 192.168.178.40.80 > 172.16.10.3.42690: Flags [..], cksum 0x290b (incorrect -> 0x3fe7), seq 124, ack 78, win 1738, options [nop,nop,TS
451 16:34:48.670161 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 14946, offset 0, flags [
452 192.168.178.201.4747 > 192.168.178.177.80: Flags [F..], cksum 0xe6f2 (incorrect -> 0x2c50), seq 107, ack 124, win 1753, options [nop,
453 16:34:48.670968 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
454 192.168.178.177.80 > 192.168.178.201.4747: Flags [..], cksum 0x2b67 (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 49
455 16:34:48.682902 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 21283, offset 0, flags [
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456 192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0xe68d (incorrect -> 0xcd78), seq 13:39, ack 56, win 1820, options [n
457 16:34:48.684073 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64088, offset 0, flags [
458 192.168.178.50.1025 > 192.168.178.201.49233: Flags [I.], cksum 0x5c15 (correct), seq 56, ack 39, win 1805, options [nop,nop,TS val 20
459 16:34:48.686884 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 64089, offset 0, flags [
460 192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0x4a25 (correct), seq 56:64, ack 39, win 1805, options [nop,nop,TS va
461 16:34:48.706109 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 21284, offset 0, flags [
462 192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0xe67b (incorrect -> 0x4a1e), seq 39:47, ack 64, win 1820, options [n
463 16:34:48.711215 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 31485, offset 0, flags
464 192.168.178.40.80 > 172.16.10.3.42704: Flags [P.], cksum 0x2985 (incorrect -> 0x61f0), seq 1:123, ack 77, win 1738, options [nop,nop
465 , length: 122
466 HTTP/1.0 200 OK
467 server: BaseHTTP/0.6 Python/3.6.8
468 date: Wed, 17 Dec 2025 08:34:48 GMT
469 content-type: text/plain
470
471 vm-2
472 16:34:48.711734 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 49063, offset 0, flags [
473 172.16.10.3.42704 > 192.168.178.40.80: Flags [I.], cksum 0x54ec (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373070
474 16:34:48.714338 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31486, offset 0, flags [
475 192.168.178.40.80 > 172.16.10.3.42704: Flags [F.], cksum 0x290b (incorrect -> 0x4e58), seq 123, ack 77, win 1738, options [nop,nop,T
476 16:34:48.714921 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 49064, offset 0, flags [
477 172.16.10.3.42704 > 192.168.178.40.80: Flags [F.], cksum 0x54e4 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537307
478 16:34:48.715094 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31487, offset 0, flags [
479 192.168.178.40.80 > 172.16.10.3.42704: Flags [I.], cksum 0x290b (incorrect -> 0x4e53), seq 124, ack 78, win 1738, options [nop,nop,TS
480 16:34:48.727378 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 4363, offset 0, flags [D
481 192.168.178.201.4759 > 192.168.178.177.80: Flags [F.], cksum 0xe6f2 (incorrect -> 0xdf14), seq 107, ack 124, win 1753, options [nop,
482 16:34:48.728924 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
483 192.168.178.177.80 > 192.168.178.201.4759: Flags [I.], cksum 0xdelf (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 49
484 16:34:48.741287 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 17501, offset 0, flags
485 192.168.178.40.80 > 172.16.10.3.42718: Flags [P.], cksum 0x2985 (incorrect -> 0xf0b8), seq 1:123, ack 77, win 1738, options [nop,nop
486 , length: 122
487 HTTP/1.0 200 OK
488 server: BaseHTTP/0.6 Python/3.6.8
489 date: Wed, 17 Dec 2025 08:34:47 GMT
490 content-type: text/plain
491
492 vm-1
493 16:34:48.741870 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 31661, offset 0, flags [
494 172.16.10.3.42718 > 192.168.178.40.80: Flags [I.], cksum 0xe196 (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373073
495 16:34:48.749243 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64090, offset 0, flags [
496 192.168.178.50.1025 > 192.168.178.201.49233: Flags [I.], cksum 0x5bad (correct), seq 64, ack 47, win 1805, options [nop,nop,TS val 20
497 16:34:48.753807 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 17502, offset 0, flags [
498 192.168.178.40.80 > 172.16.10.3.42718: Flags [F.], cksum 0x290b (incorrect -> 0xdaf8), seq 123, ack 77, win 1738, options [nop,nop,T
499 16:34:48.756342 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 31662, offset 0, flags [
500 172.16.10.3.42718 > 192.168.178.40.80: Flags [F.], cksum 0xe178 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537307
501 16:34:48.756659 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 17503, offset 0, flags [
502 192.168.178.40.80 > 172.16.10.3.42718: Flags [I.], cksum 0x290b (incorrect -> 0xdae5), seq 124, ack 78, win 1738, options [nop,nop,TS
503 16:34:48.764756 fa:16:3e:ad:d7:a2 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 15422, offset 0, flags [
504 192.168.178.201.58449 > 192.168.178.131.80: Flags [F.], cksum 0xe6c4 (incorrect -> 0xad60), seq 107, ack 124, win 1753, options [nop
505 16:34:48.765599 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 31895, offset 0, flags
506 192.168.178.40.80 > 172.16.10.3.42724: Flags [P.], cksum 0x2985 (incorrect -> 0x25ad), seq 1:123, ack 77, win 1738, options [nop,nop
507 , length: 122
508 HTTP/1.0 200 OK
509 server: BaseHTTP/0.6 Python/3.6.8
510 date: Wed, 17 Dec 2025 08:34:48 GMT
511 content-type: text/plain
512
513 vm-2
514 16:34:48.766325 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5700, offset 0, flags [D
515 172.16.10.3.42724 > 192.168.178.40.80: Flags [I.], cksum 0x1874 (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373075
516 16:34:48.766574 fa:16:3e:56:3d:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
517 192.168.178.131.80 > 192.168.178.201.58449: Flags [I.], cksum 0xac53 (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 1
518 16:34:48.772742 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31896, offset 0, flags [
519 192.168.178.40.80 > 172.16.10.3.42724: Flags [F.], cksum 0x290b (incorrect -> 0x11dc), seq 123, ack 77, win 1738, options [nop,nop,T
520 16:34:48.773274 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 5701, offset 0, flags [D
521 172.16.10.3.42724 > 192.168.178.40.80: Flags [F.], cksum 0x1863 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537307
522 16:34:48.773674 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 31897, offset 0, flags [
523 192.168.178.40.80 > 172.16.10.3.42724: Flags [I.], cksum 0x290b (incorrect -> 0x11d3), seq 124, ack 78, win 1738, options [nop,nop,TS
524 16:34:48.785375 fa:16:3e:ad:d7:a2 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 40492, offset 0, flags [
525 192.168.178.201.4769 > 192.168.178.177.80: Flags [F.], cksum 0xe6f2 (incorrect -> 0xacf3), seq 107, ack 124, win 1753, options [nop,
526 16:34:48.787409 fa:16:3e:b6:bd:46 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
527 192.168.178.177.80 > 192.168.178.201.4769: Flags [I.], cksum 0xabbc (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 49
528 16:34:48.796733 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 63757, offset 0, flags
529 192.168.178.40.80 > 172.16.10.3.42700: Flags [P.], cksum 0x2985 (incorrect -> 0xfa60), seq 1:123, ack 77, win 1738, options [nop,nop
530 , length: 122
531 HTTP/1.0 200 OK
532 server: BaseHTTP/0.6 Python/3.6.8
533 date: Wed, 17 Dec 2025 08:34:47 GMT
534 content-type: text/plain
535
536 vm-1
537 16:34:48.797070 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63758, offset 0, flags [
538 192.168.178.40.80 > 172.16.10.3.42700: Flags [F.], cksum 0x290b (incorrect -> 0xe704), seq 123, ack 77, win 1738, options [nop,nop,T
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539 16:34:48.797291 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 20976, offset 0, flags [
540     172.16.10.3.42700 > 192.168.178.40.80: Flags [.], cksum 0xeb06 (correct), seq 77, ack 123, win 58, options [nop,nop,TS val 385373078
541 16:34:48.797697 fa:16:3e:d5:b3:01 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 20977, offset 0, flags [
542     172.16.10.3.42700 > 192.168.178.40.80: Flags [F.], cksum 0xeb04 (correct), seq 77, ack 124, win 58, options [nop,nop,TS val 38537307
543 16:34:48.797783 fa:16:3e:ad:d7:a2 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63759, offset 0, flags [
544     192.168.178.40.80 > 172.16.10.3.42700: Flags [.], cksum 0x290b (incorrect -> 0xe473), seq 124, ack 78, win 1738, options [nop,nop,TS

```

amphora2的抓包结果

```

1 [root@lb-d7da4b40-1390-4922-8665-5f5e780739b8 ~]# tcpdump -i eth1 -vvv -nne
2 dropped privs to tcpdump
3 tcpdump: listening on eth1, link-type EN10MB (Ethernet), capture size 262144 bytes
4 16:34:40.965626 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 64079, offset 0, flags [
5     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe675 (incorrect -> 0x9ead), seq 771545553:771545555, ack 3316317730
6     077399232], length 2
7 16:34:40.967195 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21274, offset 0, flags [
8     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0x96a1 (correct), seq 1, ack 2, win 1820, options [nop,nop,TS val 2077
9 16:34:41.911760 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 21275, offset 0, flags [
10     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0x92e2 (correct), seq 1:3, ack 2, win 1820, options [nop,nop,TS val 2
11 16:34:41.912097 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64080, offset 0, flags [
12     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0xe673 (incorrect -> 0x8f49), seq 2, ack 3, win 1805, options [nop,nop,
13 16:34:43.965400 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 64081, offset 0, flags [
14     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe675 (incorrect -> 0x8736), seq 2:4, ack 3, win 1805, options [nop,
15 16:34:43.967916 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21276, offset 0, flags [
16     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0x7f2b (correct), seq 3, ack 4, win 1820, options [nop,nop,TS val 2077
17 16:34:44.200143 fa:16:3e:59:62:85 > ff:ff:ff:ff:ff:ff, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-ha
18 78.121, length 28
19 16:34:44.200325 fa:16:3e:59:62:85 > ff:ff:ff:ff:ff:ff, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-ha
20 78.121, length 28
21 16:34:44.913908 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 21277, offset 0, flags [
22     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0x7b6c (correct), seq 3:5, ack 4, win 1820, options [nop,nop,TS val 2
23 16:34:44.914323 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64082, offset 0, flags [
24     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0xe673 (incorrect -> 0x77d2), seq 4, ack 5, win 1805, options [nop,nop,
25 16:34:46.966597 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 64083, offset 0, flags [
26     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe675 (incorrect -> 0x6fc0), seq 4:6, ack 5, win 1805, options [nop,
27 16:34:46.968016 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21278, offset 0, flags [
28     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0x67b6 (correct), seq 5, ack 6, win 1820, options [nop,nop,TS val 2077
29 16:34:47.870802 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 59560, offset 0, flags [
30     172.16.10.3.42668 > 192.168.178.40.80: Flags [S], cksum 0x6e03 (correct), seq 2544228707, win 29200, options [mss 1460,sackOK,TS val
31 16:34:47.875770 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
32     192.168.178.40.80 > 172.16.10.3.42668: Flags [S.], cksum 0x2913 (incorrect -> 0x12a3), seq 664377209, ack 2544228708, win 27800, opt
33 24,nop,wscale 4], length 0
34 16:34:47.876254 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 19274, offset 0, flags [
35     172.16.10.3.42706 > 192.168.178.40.80: Flags [S], cksum 0xbaea (correct), seq 834073155, win 29200, options [mss 1460,sackOK,TS val
36 16:34:47.876526 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
37     192.168.178.40.80 > 172.16.10.3.42706: Flags [S.], cksum 0x2913 (incorrect -> 0x0b10), seq 2474409999, ack 834073156, win 27800, opt
38 26,nop,wscale 4], length 0
39 16:34:47.876903 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 63, id 62160, offset 0, flags [
40     172.16.10.3.42716 > 192.168.178.40.80: Flags [S], cksum 0x2913 (incorrect -> 0xfba6), seq 1278973687, win 29200, options [mss 1460,s
41 0
42 16:34:47.877056 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
43     192.168.178.40.80 > 172.16.10.3.42716: Flags [S.], cksum 0x2913 (incorrect -> 0x3c44), seq 2534247429, ack 1278973688, win 27800, op
44 127,nop,wscale 4], length 0
45 16:34:47.877170 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 59561, offset 0, flags [
46     172.16.10.3.42668 > 192.168.178.40.80: Flags [.], cksum 0xad8a (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730130 e
47 16:34:47.882928 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 59562, offset 0, flags
48     172.16.10.3.42668 > 192.168.178.40.80: Flags [P.], cksum 0x2957 (incorrect -> 0x7501), seq 1:77, ack 1, win 58, options [nop,nop,TS
49 gth: 76
50     GET / HTTP/1.1
51     Host: 172.16.10.21
52     User-Agent: curl/7.61.1
53     Accept: */*
54
55 16:34:47.883190 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 19275, offset 0, flags [
56     172.16.10.3.42706 > 192.168.178.40.80: Flags [.], cksum 0xa5f8 (correct), seq 1, ack 1, win 58, options [nop,nop,TS val 3853730131 e
57 16:34:47.883817 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 19276, offset 0, flags
58     172.16.10.3.42706 > 192.168.178.40.80: Flags [P.], cksum 0x6d70 (correct), seq 1:77, ack 1, win 58, options [nop,nop,TS val 38537301
59     GET / HTTP/1.1
60     Host: 172.16.10.21
61     User-Agent: curl/7.61.1
62     Accept: */*
63
64 16:34:47.883946 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 62161, offset 0, flags [
65     172.16.10.3.42716 > 192.168.178.40.80: Flags [.], cksum 0x290b (incorrect -> 0xd72d), seq 1, ack 1, win 58, options [nop,nop,TS val
66 16:34:47.884074 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 142: (tos 0x0, ttl 63, id 62162, offset 0, flags
67     172.16.10.3.42716 > 192.168.178.40.80: Flags [P.], cksum 0x2957 (incorrect -> 0x9ea4), seq 1:77, ack 1, win 58, options [nop,nop,TS
68 gth: 76
69     GET / HTTP/1.1
70     Host: 172.16.10.21
71     User-Agent: curl/7.61.1
72     Accept: */*
73
74 16:34:47.926415 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49437, offset 0, flags [

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75 192.168.178.40.80 > 172.16.10.3.42668: Flags [.] , cksum 0x290b (incorrect -> 0xa67c), seq 1, ack 77, win 1738, options [nop,nop,TS v
76 16:34:47.926681 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49654, offset 0, flags [
77 192.168.178.40.80 > 172.16.10.3.42716: Flags [.] , cksum 0x290b (incorrect -> 0xd01f), seq 1, ack 77, win 1738, options [nop,nop,TS v
78 16:34:47.926776 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37116, offset 0, flags [
79 192.168.178.40.80 > 172.16.10.3.42706: Flags [.] , cksum 0x290b (incorrect -> 0x9eea), seq 1, ack 77, win 1738, options [nop,nop,TS v
80 16:34:47.951422 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 63517, offset 0, flags [
81 192.168.178.50.28071 > 192.168.178.177.80: Flags [S], cksum 0xe663 (incorrect -> 0x09ad), seq 390457353, win 28040, options [mss 140
82 th 0
83 16:34:47.954187 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
84 192.168.178.177.80 > 192.168.178.50.28071: Flags [S.], cksum 0x3cf0 (correct), seq 3431626439, ack 390457354, win 27800, options [ms
85 scale 4], length 0
86 16:34:47.955067 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63518, offset 0, flags [
87 192.168.178.50.28071 > 192.168.178.177.80: Flags [.] , cksum 0xe65b (incorrect -> 0xd13a), seq 1, ack 1, win 1753, options [nop,nop,T
88 16:34:47.964604 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 58259, offset 0, flags [
89 192.168.178.50.51301 > 192.168.178.131.80: Flags [S], cksum 0xe635 (incorrect -> 0xd0df), seq 972059992, win 28040, options [mss 140
90 h 0
91 16:34:47.974311 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 1946, offset 0, flags [D
92 192.168.178.50.28079 > 192.168.178.177.80: Flags [S], cksum 0xe663 (incorrect -> 0x49fa), seq 3232497206, win 28040, options [mss 14
93 gth 0
94 16:34:47.975592 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
95 192.168.178.177.80 > 192.168.178.50.28079: Flags [S.], cksum 0x804d (correct), seq 2649662013, ack 3232497207, win 27800, options [m
96 wscale 4], length 0
97 16:34:47.976154 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 1947, offset 0, flags [D
98 192.168.178.50.28079 > 192.168.178.177.80: Flags [.] , cksum 0xe65b (incorrect -> 0x149a), seq 1, ack 1, win 1753, options [nop,nop,T
99 16:34:47.976553 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
100 192.168.178.131.80 > 192.168.178.50.51301: Flags [S.], cksum 0xb3b9 (correct), seq 1867301045, ack 972059993, win 27800, options [ms
101 cale 4], length 0
102 16:34:47.976744 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 58260, offset 0, flags [
103 192.168.178.50.51301 > 192.168.178.131.80: Flags [.] , cksum 0xe62d (incorrect -> 0x47fc), seq 1, ack 1, win 1753, options [nop,nop,T
104 16:34:47.997045 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 63519, offset 0, flags
105 192.168.178.50.28071 > 192.168.178.177.80: Flags [P.], cksum 0xe6c5 (incorrect -> 0xdc36), seq 1:107, ack 1, win 1753, options [nop,
106 TTP, length: 106
107 GET / HTTP/1.1
108 host: 172.16.10.21
109 user-agent: curl/7.61.1
110 accept: */*
111 x-forwarded-for: 172.16.10.3
112
113 16:34:47.998456 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 23935, offset 0, flags [
114 192.168.178.177.80 > 192.168.178.50.28071: Flags [.] , cksum 0xd089 (correct), seq 1, ack 107, win 1738, options [nop,nop,TS val 4082
115 16:34:48.004107 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 58261, offset 0, flags
116 192.168.178.50.51301 > 192.168.178.131.80: Flags [P.], cksum 0xe697 (incorrect -> 0x5306), seq 1:107, ack 1, win 1753, options [nop,
117 TP, length: 106
118 GET / HTTP/1.1
119 host: 172.16.10.21
120 user-agent: curl/7.61.1
121 accept: */*
122 x-forwarded-for: 172.16.10.3
123
124 16:34:48.007912 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 172: (tos 0x0, ttl 64, id 1948, offset 0, flags [
125 192.168.178.50.28079 > 192.168.178.177.80: Flags [P.], cksum 0xe6c5 (incorrect -> 0x1fa1), seq 1:107, ack 1, win 1753, options [nop,
126 TTP, length: 106
127 GET / HTTP/1.1
128 host: 172.16.10.21
129 user-agent: curl/7.61.1
130 accept: */*
131 x-forwarded-for: 172.16.10.3
132
133 16:34:48.008686 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 46335, offset 0, flags [
134 192.168.178.131.80 > 192.168.178.50.51301: Flags [.] , cksum 0xe62d (incorrect -> 0x4764), seq 1, ack 107, win 1738, options [nop,nop
135 16:34:48.009313 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 33964, offset 0, flags [
136 192.168.178.177.80 > 192.168.178.50.28079: Flags [.] , cksum 0x13ff (correct), seq 1, ack 107, win 1738, options [nop,nop,TS val 4082
137 16:34:48.069984 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 33965, offset 0, flags
138 192.168.178.177.80 > 192.168.178.50.28079: Flags [P.], cksum 0xe5b5 (correct), seq 1:118, ack 107, win 1738, options [nop,nop,TS val
139 h: 117
140 HTTP/1.0 200 OK
141 Server: BaseHTTP/0.6 Python/3.6.8
142 Date: Wed, 17 Dec 2025 08:34:47 GMT
143 Content-Type: text/plain
144
145 16:34:48.070206 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 1949, offset 0, flags [D
146 192.168.178.50.28079 > 192.168.178.177.80: Flags [.] , cksum 0xe65b (incorrect -> 0x12ff), seq 107, ack 118, win 1753, options [nop,n
147 16:34:48.074007 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 33966, offset 0, flags [
148 192.168.178.177.80 > 192.168.178.50.28079: Flags [P.], cksum 0x6560 (correct), seq 118:123, ack 107, win 1738, options [nop,nop,TS v
149 16:34:48.074241 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 1950, offset 0, flags [D
150 192.168.178.50.28079 > 192.168.178.177.80: Flags [.] , cksum 0xe65b (incorrect -> 0x12f5), seq 107, ack 123, win 1753, options [nop,n
151 16:34:48.088047 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 49655, offset 0, flags
152 192.168.178.40.80 > 172.16.10.3.42716: Flags [P.], cksum 0x2980 (incorrect -> 0x8111), seq 1:118, ack 77, win 1738, options [nop,nop
153 , length: 117
154 HTTP/1.0 200 OK
155 server: BaseHTTP/0.6 Python/3.6.8
156 date: Wed, 17 Dec 2025 08:34:47 GMT
157 content-type: text/plain
```



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158
159 16:34:48.088376 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 62163, offset 0, flags [
160 172.16.10.3.42716 > 192.168.178.40.80: Flags [.], cksum 0x290b (incorrect -> 0xd4c6), seq 77, ack 118, win 58, options [nop,nop,TS v
161 16:34:48.097310 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 64084, offset 0, flags [
162 192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe68d (incorrect -> 0xfbf6), seq 6:32, ack 5, win 1805, options [nop
163 16:34:48.098688 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21279, offset 0, flags [
164 192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0x5ec6 (correct), seq 5, ack 32, win 1820, options [nop,nop,TS val 207
165 16:34:48.106737 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 23936, offset 0, flags
166 192.168.178.177.80 > 192.168.178.50.28071: Flags [P.], cksum 0xa211 (correct), seq 1:118, ack 107, win 1738, options [nop,nop,TS val
167 h: 117
168 HTTP/1.0 200 OK
169 Server: BaseHTTP/0.6 Python/3.6.8
170 Date: Wed, 17 Dec 2025 08:34:47 GMT
171 Content-Type: text/plain
172
173 16:34:48.107126 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63520, offset 0, flags [
174 192.168.178.50.28071 > 192.168.178.177.80: Flags [.], cksum 0xe65b (incorrect -> 0xcf2b), seq 107, ack 118, win 1753, options [nop,n
175 16:34:48.107266 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 183: (tos 0x0, ttl 64, id 46336, offset 0, flags
176 192.168.178.131.80 > 192.168.178.50.51301: Flags [P.], cksum 0xe6a2 (incorrect -> 0x18f4), seq 1:118, ack 107, win 1738, options [no
177 HTTP, length: 117
178 HTTP/1.0 200 OK
179 Server: BaseHTTP/0.6 Python/3.6.8
180 Date: Wed, 17 Dec 2025 08:34:47 GMT
181 Content-Type: text/plain
182
183 16:34:48.107331 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 58262, offset 0, flags [
184 192.168.178.50.51301 > 192.168.178.131.80: Flags [.], cksum 0xe62d (incorrect -> 0x4615), seq 107, ack 118, win 1753, options [nop,n
185 16:34:48.108744 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 23937, offset 0, flags [
186 192.168.178.177.80 > 192.168.178.50.28071: Flags [P.], cksum 0x218c (correct), seq 118:123, ack 107, win 1738, options [nop,nop,TS v
187 16:34:48.109293 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63521, offset 0, flags [
188 192.168.178.50.28071 > 192.168.178.177.80: Flags [.], cksum 0xe65b (incorrect -> 0xcfc22), seq 107, ack 123, win 1753, options [nop,n
189 16:34:48.109396 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 23938, offset 0, flags [
190 192.168.178.177.80 > 192.168.178.50.28071: Flags [F.], cksum 0xcf31 (correct), seq 123, ack 107, win 1738, options [nop,nop,TS val 4
191 16:34:48.112037 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 46337, offset 0, flags [
192 192.168.178.131.80 > 192.168.178.50.51301: Flags [P.], cksum 0xe632 (incorrect -> 0x9872), seq 118:123, ack 107, win 1738, options [
193 HTTP
194 16:34:48.112147 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 58263, offset 0, flags [
195 192.168.178.50.51301 > 192.168.178.131.80: Flags [.], cksum 0xe62d (incorrect -> 0x4605), seq 107, ack 123, win 1753, options [nop,n
196 16:34:48.116082 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 33967, offset 0, flags [
197 192.168.178.177.80 > 192.168.178.50.28079: Flags [F.], cksum 0x12d6 (correct), seq 123, ack 107, win 1738, options [nop,nop,TS val 4
198 16:34:48.119043 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 71: (tos 0x0, ttl 64, id 49656, offset 0, flags [
199 192.168.178.40.80 > 172.16.10.3.42716: Flags [P.], cksum 0x2910 (incorrect -> 0x206b), seq 118:123, ack 77, win 1738, options [nop,n
200 16:34:48.119269 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 62164, offset 0, flags [
201 172.16.10.3.42716 > 192.168.178.40.80: Flags [.], cksum 0x290b (incorrect -> 0xd483), seq 77, ack 123, win 58, options [nop,nop,TS v
202 16:34:48.136182 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49657, offset 0, flags [
203 192.168.178.40.80 > 172.16.10.3.42716: Flags [F.], cksum 0x290b (incorrect -> 0xcde1), seq 123, ack 77, win 1738, options [nop,nop,T
204 16:34:48.136606 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 62165, offset 0, flags [
205 172.16.10.3.42716 > 192.168.178.40.80: Flags [F.], cksum 0x290b (incorrect -> 0xd45e), seq 77, ack 124, win 58, options [nop,nop,TS
206 16:34:48.137109 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49658, offset 0, flags [
207 192.168.178.40.80 > 172.16.10.3.42716: Flags [.], cksum 0x290b (incorrect -> 0xcdce), seq 124, ack 78, win 1738, options [nop,nop,TS
208 16:34:48.151422 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63522, offset 0, flags [
209 192.168.178.50.28071 > 192.168.178.177.80: Flags [.], cksum 0xe65b (incorrect -> 0xcef6), seq 107, ack 124, win 1753, options [nop,n
210 16:34:48.157121 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 1951, offset 0, flags [D
211 192.168.178.50.28079 > 192.168.178.177.80: Flags [.], cksum 0xe65b (incorrect -> 0x1274), seq 107, ack 124, win 1753, options [nop,n
212 16:34:48.161313 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 46338, offset 0, flags [
213 192.168.178.131.80 > 192.168.178.50.51301: Flags [F.], cksum 0xe62d (incorrect -> 0x45e2), seq 123, ack 107, win 1738, options [nop,
214 16:34:48.168104 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 68: (tos 0x0, ttl 64, id 21280, offset 0, flags [
215 192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0x5e73 (correct), seq 5:7, ack 32, win 1820, options [nop,nop,TS val
216 16:34:48.169022 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64085, offset 0, flags [
217 192.168.178.50.1025 > 192.168.178.201.49233: Flags [F.], cksum 0xe673 (incorrect -> 0x5e47), seq 32, ack 7, win 1805, options [nop,no
218 16:34:48.191723 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 1952, offset 0, flags [D
219 192.168.178.50.28079 > 192.168.178.177.80: Flags [F.], cksum 0xe65b (incorrect -> 0x1251), seq 107, ack 124, win 1753, options [nop,
220 16:34:48.193408 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
221 192.168.178.177.80 > 192.168.178.50.28079: Flags [.], cksum 0x1213 (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 40
222 16:34:48.201980 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 49438, offset 0, flags
223 192.168.178.40.80 > 172.16.10.3.42668: Flags [P.], cksum 0x2985 (incorrect -> 0xb748), seq 1:123, ack 77, win 1738, options [nop,nop
224 , length: 122
225 HTTP/1.0 200 OK
226 server: BaseHTTP/0.6 Python/3.6.8
227 date: Wed, 17 Dec 2025 08:34:47 GMT
228 content-type: text/plain
229
230 vm-2
231 16:34:48.202195 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 58264, offset 0, flags [
232 192.168.178.50.51301 > 192.168.178.131.80: Flags [.], cksum 0xe62d (incorrect -> 0x4579), seq 107, ack 124, win 1753, options [nop,n
233 16:34:48.203447 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 59563, offset 0, flags [
234 172.16.10.3.42668 > 192.168.178.40.80: Flags [.], cksum 0x290b (incorrect -> 0xaa38), seq 77, ack 123, win 58, options [nop,nop,TS v
235 16:34:48.211164 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49439, offset 0, flags [
236 192.168.178.40.80 > 172.16.10.3.42668: Flags [F.], cksum 0x290b (incorrect -> 0xa39d), seq 123, ack 77, win 1738, options [nop,nop,T
237 16:34:48.211520 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 59564, offset 0, flags [
238 172.16.10.3.42668 > 192.168.178.40.80: Flags [F.], cksum 0x290b (incorrect -> 0xaa22), seq 77, ack 124, win 58, options [nop,nop,TS
239 16:34:48.211696 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 49440, offset 0, flags [
240 192.168.178.40.80 > 172.16.10.3.42668: Flags [.], cksum 0x290b (incorrect -> 0xa392), seq 124, ack 78, win 1738, options [nop,nop,TS
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241 16:34:48.218432 fa:16:3e:09:88:25 > fa:16:3e:b6:bd:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 63523, offset 0, flags [
242     192.168.178.50.28071 > 192.168.178.177.80: Flags [F.], cksum 0xe65b (incorrect -> 0xcceb2), seq 107, ack 124, win 1753, options [nop,
243 16:34:48.220374 fa:16:3e:b6:bd:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
244     192.168.178.177.80 > 192.168.178.50.28071: Flags [.], cksum 0xcxe52 (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 40
245 16:34:48.222948 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 188: (tos 0x0, ttl 64, id 37117, offset 0, flags
246     192.168.178.40.80 > 172.16.10.3.42706: Flags [P.], cksum 0x2985 (incorrect -> 0xb0a3), seq 1:123, ack 77, win 1738, options [nop,nop
247 , length: 122
248     HTTP/1.0 200 OK
249     server: BaseHTTP/0.6 Python/3.6.8
250     date: Wed, 17 Dec 2025 08:34:47 GMT
251     content-type: text/plain
252
253     vm-1
254 16:34:48.223300 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 19277, offset 0, flags [
255     172.16.10.3.42706 > 192.168.178.40.80: Flags [.], cksum 0x290b (incorrect -> 0xa27e), seq 77, ack 123, win 58, options [nop,nop,TS v
256 16:34:48.229667 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 64086, offset 0, flags [
257     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe68d (incorrect -> 0xf5ad), seq 32:58, ack 7, win 1805, options [no
258 16:34:48.230544 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37118, offset 0, flags [
259     192.168.178.40.80 > 172.16.10.3.42706: Flags [F.], cksum 0x290b (incorrect -> 0x9be5), seq 123, ack 77, win 1738, options [nop,nop,T
260 16:34:48.230779 fa:16:3e:d5:b3:01 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 63, id 19278, offset 0, flags [
261     172.16.10.3.42706 > 192.168.178.40.80: Flags [F.], cksum 0x290b (incorrect -> 0xa26c), seq 77, ack 124, win 58, options [nop,nop,TS
262 16:34:48.231012 fa:16:3e:09:88:25 > fa:16:3e:d5:b3:01, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 37119, offset 0, flags [
263     192.168.178.40.80 > 172.16.10.3.42706: Flags [.], cksum 0x290b (incorrect -> 0x9bdb), seq 124, ack 78, win 1738, options [nop,nop,TS
264 16:34:48.231443 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 21281, offset 0, flags [
265     192.168.178.201.49233 > 192.168.178.50.1025: Flags [.], cksum 0x5da2 (correct), seq 7, ack 58, win 1820, options [nop,nop,TS val 207
266 16:34:48.237589 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 21282, offset 0, flags [
267     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0x4bd5 (correct), seq 7:15, ack 58, win 1820, options [nop,nop,TS val
268 16:34:48.237977 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64087, offset 0, flags [
269     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0xe673 (incorrect -> 0x5d9a), seq 58, ack 15, win 1805, options [nop,n
270 16:34:48.241552 fa:16:3e:09:88:25 > fa:16:3e:56:3d:46, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 58265, offset 0, flags [
271     192.168.178.50.51301 > 192.168.178.131.80: Flags [F.], cksum 0xe62d (incorrect -> 0x4551), seq 107, ack 124, win 1753, options [nop,
272 16:34:48.255402 fa:16:3e:56:3d:46 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 0, offset 0, flags [DF],
273     192.168.178.131.80 > 192.168.178.50.51301: Flags [.], cksum 0x4505 (correct), seq 124, ack 108, win 1738, options [nop,nop,TS val 12
274 16:34:48.418773 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 92: (tos 0x0, ttl 64, id 21283, offset 0, flags [
275     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0xcd78 (correct), seq 15:41, ack 58, win 1820, options [nop,nop,TS va
276 16:34:48.419066 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64088, offset 0, flags [
277     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0xe673 (incorrect -> 0x5c15), seq 58, ack 41, win 1805, options [nop,n
278 16:34:48.421772 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 64089, offset 0, flags [
279     192.168.178.50.1025 > 192.168.178.201.49233: Flags [P.], cksum 0xe67b (incorrect -> 0x4a25), seq 58:66, ack 41, win 1805, options [n
280 16:34:48.442082 fa:16:3e:ad:d7:a2 > fa:16:3e:09:88:25, ethertype IPv4 (0x0800), length 74: (tos 0x0, ttl 64, id 21284, offset 0, flags [
281     192.168.178.201.49233 > 192.168.178.50.1025: Flags [P.], cksum 0x4a1e (correct), seq 41:49, ack 66, win 1820, options [nop,nop,TS va
282 16:34:48.484167 fa:16:3e:09:88:25 > fa:16:3e:ad:d7:a2, ethertype IPv4 (0x0800), length 66: (tos 0x0, ttl 64, id 64090, offset 0, flags [
283     192.168.178.50.1025 > 192.168.178.201.49233: Flags [.], cksum 0xe673 (incorrect -> 0x5bad), seq 66, ack 49, win 1805, options [nop,n
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